Name_____ Date_____ Period_____

**LESSON 2-4 · HOMEWORK PACKET · SHEET 1 OF 2**

# Building a Box Plot

Maryland Standard 6.SP.B.4 · Learning target – I can find the five-number summary of a data set and use it to build a box plot.

**REMEMBER**

- **Five-number summary**: minimum, Q1, median, Q3, maximum.
- **Q1** is the median of the lower half; **Q3** of the upper half. Odd count: leave the median out.
- The **box** goes from Q1 to Q3, with a line at the median. **Whiskers** reach the min and max.

**WORKED EXAMPLE**

Find the five-number summary of 4, 6, 9, 10, 13, 15, 18.

**Step 1** Min = 4, max = 18, median = 10

**Step 2** Lower half 4, 6, 9, so Q1 = 6

**Step 3** Upper half 13, 15, 18, so Q3 = 15

**Answer** **4, 6, 10, 15, 18**

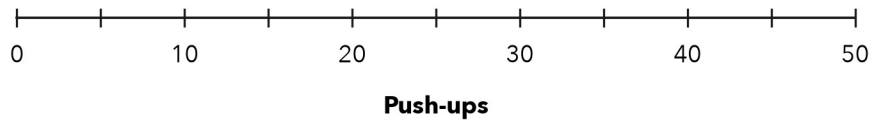
## PRACTICE

*show your work · label units*

**1.** **Push-ups.** Nine students did 22, 40, 10, 34, 18, 46, 25, 14, and 30 push-ups. Find the five-number summary.

| Min | Q1 | Median | Q3 | Max |
|---|---|---|---|---|
| | | | | |

**2.** Draw a box plot of the push-up data from Problem 1.



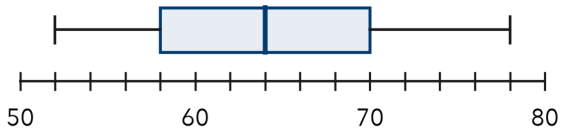

**Push-ups**

**3.** **Temperatures.** The box plot shows daily high temperatures (°F) for a month.

**a.** Median: _____ °F

**b.** Q1 and Q3: _____

**c.** Fraction of days from 58 to 70°F: _____




**Temperature (°F)**

**4.** **Homework Minutes.** 26, 14, 39, 20, 11, 32, 18, 28. Find each value.

**a.** Q1: ____ **b.** Median: ____ **c.** Q3: ____

**5.** **Find the Error.** Grace says Q1 of 2, 5, 7, 8, 12, 14, 20 is 6. She used 2, 5, 7, 8 as the lower half. Explain her error and find Q1.

_____

_____

| CHECK YOURSELF | ☐ I can do this on my own | ☐ I need a little help | ☐ I have a question about #___ |
|---|---|---|---|

**AT HOME** Explain to someone at home what the box in a box plot shows.

Name_____ Date_____ Period_____

**LESSON 2-4 · HOMEWORK PACKET · SHEET 2 OF 2**

# Comparing Box Plots

Maryland Standard 6.SP.B.4 · Learning target – I can compare two groups by reading the medians and box widths of their box plots.

**REMEMBER**

- Compare **centers**: use the medians.
- Compare **spread**: box width and range.
- A **narrower box** (Q3 − Q1) is more **consistent**.

**WORKED EXAMPLE**

A: median 18, box 14–22. B: median 24, box 20–26.

**Step 1** Center: 24 > 18, so B scored higher.

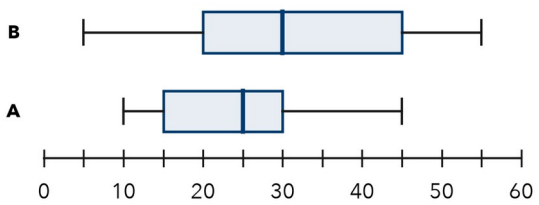
**Step 2** Box widths: A is 22 − 14 = 8; B is 26 − 20 = 6.

**Answer** **B is higher and more consistent**

## PRACTICE

*show your work · label units*

**1. Reading Time.** Minutes of reading per day for Class A and Class B.




**a.** Greater median: ____ by ____ min

**b.** Box widths: A ____ B ____

**c.** More consistent: _____

**2.** Find each measure of spread.

**a.** Q1 = 12, Q3 = 30. Box width: _____

**b.** Min = 5, max = 41. Range: _____

**3. Bowling.** Complete the table for two bowlers' scores.

| | Min | Q1 | Median | Q3 | Max | Box width | Range |
|---|---|---|---|---|---|---|---|
| **Maya** | 80 | 95 | 102 | 110 | 125 | | |
| **Leon** | 70 | 90 | 105 | 112 | 140 | | |

**Who is more consistent?**_____

**4. Plant Heights.** Both medians are 14 cm. Box S: 12–16 cm. Box T: 8–20 cm. Which varies more?

☐ Group S ☐ Group T

**5. Choose a Bus.** Both medians are 20 min. Boxes: A 18–22, B 14–26 min. Which is more reliable?

_____

## TEST PRACTICE

*choose the best answer*

**6.** X: Q1 = 22, Q3 = 34. Y: Q1 = 25, Q3 = 31. Which middle half is more consistent?

**A** X **B** Y **C** They are equal **D** Cannot tell

**7.** A box plot has the summary 8, 12, 15, 21, 30. How wide is the box?

**A** 9 **B** 22 **C** 15 **D** 3

| | | | |
|---|---|---|---|
| **CHECK YOURSELF** | □ I can do this on my own | □ I need a little help | □ I have a question about #___ |